

Interview Experience-Batch 10

PGDBA – Batch 10

INTERVIEW

EXPERIENCE

Alumni Team- Batch 11

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QRT

Sachin Goyal

Written test (1 hr):

1. 10 Questions about basic stats,
2. 10 Questions about pandas, python code (given a function what it will return or give syntax error, these kind of basic questions)
3. 10 Questions about Finance.(I had no knowledge about these questions, so I just took a random guess for every question)

Round 1 (25-30 mins):

1. Basic Regression Questions like multicollinearity, Lasso and Ridge, left and Right skewed distribution difference, All the regression model assumptions etc.
2. Why do you want to join QRT?
3. What was your role in the previous company and why did you leave?
4. Case: How will you predict the Corn Price, what will be the features you will consider?

(The last question was almost asked to everyone, but with changed commodities, someone was asked about rice, wheat, crude oil, metal etc. All these kinds of questions share a similar approach/Framework. You need to consider Demand and Supply side features, Macroeconomic features, and some other factors. You can ask GPT about it about the Framework)

Round 2 (2-3 mins):

1. It was a rapid-fire round.

2. I was working in the defence domain before coming to PGDBA, so they just asked me about what the Lnt Defence model is to earn money and what features will be helpful to determine the Defence industry growth.

Round 3 (30-40mins): (Completely Finance Based)

1. It was an online round in which a guy from Europe joined in.
2. I was given a case/situation:
Suppose there were only 2 markets (China & US) for trading in the whole world and one commodity is being traded in both the markets at the same price and in China market supply is lower than demand and in the US market supply is higher than demand.
3. So, do you think Trade should happen between these two markets if the price of commodities is same in both markets?
4. At what price differential, do you think the trade between the market will start?
5. So, what do you think will happen in future, will this trend continue, the price of commodities change or not?
6. What will be your investing strategy and at what level of risk you will pull out of the trade?

Jahnavi Kalisetty

QRT checks your thought process and your ability to apply concepts to solve problems rather than the concepts per se and it's a holistic evaluation

Test Round:

This round consisted of questions of Python (basic python + Pandas). Python questions are on Yield and pandas are mostly on group by and filling missing values using different methods(bfill() and ffill()) – I would suggest whatever project you do, try not to use chatgpt for basic stuff esp. EDA work, This helps in clearing test rounds wherever there is coding in basic concepts

Stats and ML: one on proportion test and other basic ML like Type1 error, Type2 error. Correlation range, Ridge or lasso or elasticnet for both interpretability and overfitting. All ML questions are doable if we study basic regression and other ML concepts

Finance: About some exchanges and what is the most traded commodity there. And application oriented like when will the crude oil/gold ratio change.

Round 1: (About 40mins)

14 people got shortlisted and 3 were waitlisted (They took interview of all 17 people)

Personality Based:

1. About my intro, I have background of UPSC, so questions based on that
2. I changed company after working for a year, so he asked about that (There might be profile-based questions, to check your personality, so if your profile has any hiccups like frequent company changes or UPSC prep, be thorough about what you're going to answer)

Technical:

1. Trading strategies, Mean Reverting and Pairs trading. And out of all the universe of stocks you have how do you select pairs (here they were looking for strategy). And follow up questions on these because I said I'll choose pairs based on covariance and cointegration, what is cointegration. In the pair of stocks with increasing variance do you think covariance works to find out pairs. So there cointegration helps. (Interviewer was directing me to think in the direction of co integration by giving hints, It was more of a discussion than question and answer based and 90% of my interview is on these trading strategies and applying concepts of time series and basics like variance-covariance to these strategies)
2. Can you name few statistical tests which you know of
3. Of the tests you mentioned can you think of any test which you can apply for the above mean reversion strategy to check the significance of the mean reverting signal
4. Difference between R^2 and Adj R^2
5. Methods to address overfitting
6. Difference between ridge and lasso and when will you use which one
7. What is Type1 and Typer 2 error
8. What is stationarity and ADF test
9. I would like to invest in the FMCG sector, let's say in HUL at some point of time in the near future, how should my strategy be? (Here they were looking for signals)

Most of the discussion was on trading strategies and how we can apply our knowledge of statistics to validate our signals/hypothesis

Round 2: (About 10 mins)

Quite unconventional round, they gave me a problem statement and expected me to solve it, and they were open for discussion, you can think out loud and tell them what strategies you are thinking of and if they have any different strategy in their mind, they'll guide you to that by asking relevant questions

We have natural gas consumption pattern for the past 51 years and in the past there was some pattern and the same pattern repeated later at some point of time and you want to detect that pattern, basically he asked to check if there is any repeating pattern in your time series data, how do you check that (This is not cyclicity and they won't come in cycles or seasons, just at a random point of time there was a pattern repetition)

Round 3: (About 20-25 mins)

(Two panels, one for Energy and one for Metals)

This is more of a cultural fit round

1. Tell me about yourself
2. Why QRT
3. Why did you join PGDBA after working as senior data scientist
4. What were your day-to-day activities in your last job as senior data scientist
5. Tell me about any project you have worked on, as a Data Scientist
6. Were you alone in doing this project or you worked in a team and what's your role in that team
7. Tell me about your city (This was common question for almost all of us in that panel)
8. Tell me about your family, Hobbies
9. Tell me about organic farming (hobby). Try to keep interesting hobbies so that they get curious about it

Suggestions:

1. Practice coding basic, pandas and python although we (Batch 10) didn't have DSA in our test, we never know what QRT tests on for your batch. So, start practicing DSA too. And don't use chatgpt for everything, you'll be doing lot of projects in KGP, get hands on esp., in EDA and fitting simple models
2. Try to do FRM project something relating to trading strategies which will help you in knowing and understanding those strategies better, whether you keep them in your resume or not as knowing them helps in Hedge Fund companies
3. Finance is important, know about derivatives and what are the different instruments in derivatives, and which one do you use in which situation, mostly application of the things that you know of.
4. FRM and CRF are important if you are targeting Quant companies. SAAPM is important if you want in depth understanding of Time series and which is necessary for Quant companies. For better understanding of Regression, Econometrics can be audited or can be taken as elective
5. Try to develop mathematical understanding of regression from RTSM in IIT KGP and application oriented from SAAPM and Econometrics

Reeshav Samanta

Written Test (Proctored – in-venue exam):

30 questions were to be answered in 60 min. 10 questions were on Statistics, Inference, Basic ML, another 10 on Programming: Output of the code (Basic DSA), Numpy, Pandas, Dataframe Manipulation etc., last 10 on Commodities Trading, Economics, Basic Finance, Import-Export etc. All the questions had 4 options, hence option elimination was my approach.

Round-1 (Face to face interview)

1. Introduce yourself (talked about my work-ex at Aditya Birla Group, Hindalco)
2. Explain in detail about the role – talked about my role in Smelter Cast House.
3. Explain the end-to-end value chain of Aluminium – told the journey of Bauxite in mines to Alumina to Al in Smelter to Cast House then to downstream.
4. Asked me to explain the products of Alu. Answered the downstream products & their uses. He asked if I can explain the Smelter in detail – all the parts like Carbon, Cast House, Potline, did so.
5. Who is the biggest Alu. producer in the world & India – answered Rio Tinto Alco & Vedanta Aluminium (by volume) & Hindalco Industries Ltd. (by diversity of products)
6. Is India a net exporter or net importer of Alu. – answered net importer – he asked why do I feel so – as I felt India still depends on Middle Eastern smelters for high grade Alu.
7. Asked me to explain my time-series project – It was on IP of electricity in US – he asked if I knew the meaning of IP (Industrial Prod. Index) & where

is that used – I answered but was not clear – then I continued explaining the time series project & approach taken.

8. Asked if I can answer some questions on Business Analytics – I said yes – He asked what is stationarity – answered

9. Why is log-transformation used? I tried to explain through the lens of regression & time-series but he asked me to keep it simple – answered.

10. Asked a puzzle -

<https://www.geeksforgeeks.org/aptitude/measuring-6l-water-4l-9l-buckets/> (Partially answered – tried but unable to recall the exact solution)

11. On which indices Alu. Is traded – knew about LME, told that.

12. How would you predict the yield of a bauxite mine – I first couldn't get the exact question as I thought it's an un-commissioned mine – later the interviewer re-framed the question to which I tried answering like using drill reports, weather data, soil content, last year's yield.

Round-2 (Face to face interview)

1. How would you determine fair price of Cotton – explained the demand & supply side factors, macroeconomic factors like currency exchange etc. The interviewer asked me to forget about reg., time-series, ML etc. for a while & just tell the approach & factors.

2. What alternate datasets would you use – told something about soil reports & shipments data.

3. Why QRT after ABG? Told about my passion for commodities as a mechanical engg. at ABG (dealt HFO, Propane, Argon, metals etc.) & my desire to learn the financial part of it.

This was a short interview.

Round-3 (Online interview with Senior Research Director)

Directly asked me a question.

Suppose there are two countries in this world – US & China – US has a surplus of commodities inventory & Chn with a deficit. How would you create a trading strategy if you have this situation?

This was a tricky question where I had to frame a question to myself. I explained by saying about the supply demand curve in both places – Prices going up in China while down in the US. First explained imp arb – buying physical inventory in the US & selling in China. However, he said a financial trader won't do so. (Obviously!!)

He asked me to think deeper in terms of Futures. I said something related to that. He asked how the situation is if there are two exchanges v/s single exchange. Hinted me to arbitrages. Asked how I would think about the dynamics of production in both the countries.

In between he asked me if I did my bachelors in Economics to which I denied. He asked if I studied Eco. in my post grad, to which I explained to him that my majors are in Business Analytics.

Finally, he asked as an analyst what all way I can model this situation – I spoke about Currency Exchange, Import Export volumes. He hinted me to think about import tariffs & CIF (freight charges etc.)

He asked me if I had any question – Asked him how much he likes the multinational global culture at QRT. He was kind enough to answer it.

Tips:

1. Ask for time in case you have to think for a question. Helps you to frame your answer.
2. Seek clarification before answering if you have doubt with the question.

3. In general, it's better to answer a question in case you have little knowledge about that, but also giving a disclaimer: "Allow me to make a guess. I feel so because...."

Soham Mittal

Online test - Came for energy team (2 ppl) and metals team (2 ppl). Test + CV based shortlisting. Preferred people with energy or metals work ex.

The test had pandas/numpy, probability/stats and finance ques. Selected 14ppl + 3ppl waitlist for round 1.

Yoogesh Kumar M

Round1:

1. Tell me about yourself.
2. Explain your regression project
3. How did you handle multicollinearity
4. Case: India is importing metal from 5 countries, forecast how much we should import from each country.
5. Puzzle: How to measure 4L with 5 L and 3L jar

Round 2:

1. Why QRT.
2. Case: Find if the Cotton Price USA is selling is fare
3. What is Type 1 and Type 2 Error.

Niranjan Herwadkar

Round 1:

1. Discussion on my SARIMA project
2. Case study: You are an aluminium producer in India; will you sell your produce to Germany or to China? What factors will you consider?
3. What is adjusted R^2 , and why do we need it?

Round 2:

1. Questions on my background
2. Motivation to join QRT
3. Some other personal/HR questions

Round 3:

It was conducted with the head trader based in Paris.

1. Asked me to explain my BTech internship project (since it was relevant to trading). There were many follow-up questions, like – How is natural gas stored?
2. How good are you at coding?

Piyus Pramanik

1. Puzzles like in quant interviews
2. Questions about commodities and business
3. What are the different factors affecting prices of commodities for example cotton
4. How do you estimate price of cotton

Second round

1. How do you think Hindalco is selling aluminum at this price, what goes into pricing
2. What is the business model used by Maruti Suzuki to be profitable

DE SHAW

Aniket Thakur

1. Questions about the case they had given prior to the interview
2. They cross questioned about the features considered, features rejected, the interaction terms used in the project etc. A clear intent and proper justification was expected
3. Assumptions of Central Limit theorem
4. What tests to conclude normality of variables was used in the case? They had used the Shapiro- wilk test and were questioned about why it was used and not any other etc.
5. What are exogenous variables?
6. Multiple cross questioning about how do you know if the applied model is good for data, and if there is some remaining pattern in the data after modelling, how to check these?
7. What are restricted and unrestricted models in ANOVA? How do you check for the statistical significance of the coefficients
8. What is multi collinearity and how to address it?

Second round

1. Questions on time series
2. Multiple cross questions about time series as applied to the project, what are exogenous variables, what kind of exogenous variables can be applied here? Do you estimate everything again, or do you keep the original + exogenous ? Does it make a difference??
3. They shared a screen and asked a finance question, variance and co variance between companies, what kind of portfolio should you have?

Questions about NLP based on the project he had done: What are glove embeddings, how are they different from word to vec? And other basic questions about analysing performance metrics etc.

Discussion on Car price prediction model project:

1. Questions about why exactly these features are considered and why not more?
2. The interviewer suggested some features like fuel price, and possibility of a recent model of car being released etc. He had to justify the reasons behind not using these
3. Discussion about ridge and lasso regression, and their closed forms and why lasso results in deletion of some variables?

Final round

They gave some questions and asked to solve as many as he could

1. Derive coefficients for simple linear regression
2. Prove that projection matrix is idempotent
3. Pseudo code to check if a given number is prime or not
4. Question about risk management
5. A scenario was given with students and their marks distribution and there were missing values and the discussion was about how would one go about addressing these.

Soham mittal

DCL (1 week) - Team of 2 ppl. Case comp. The goal was to collect data and predict some house price index for counties in the USA. We had to submit data collected, code and ppt. Online leaderboard will update every day. Focus both on accuracy as well as interpretability.

PPT round (40 mins) - 21 teams were selected for this round. They focused a lot on data collected, feature engineering, feature selection and what more could have been done. Some basic ML theory questions (normality tests, how to remove multicollinearity, VIF) related to what we had done were also asked.

Round 1 (40 mins) - ***27 people were selected for this round. I***

1. Introduction- Favorite project, deep dived on that. Asked a few questions from other projects as well. Make sure to know each keyword in your resume inside out. Some questions on projects - how to handle class imbalance, diff b/w correlation and cointegration, what is FEVD.
2. Asked a case - Given 2 locations (delhi, hyderabad), you have to choose one to set up a hotel. What factors will you take into account?
3. Asked a puzzle - Torch and Bridge puzzle from GFG.
4. More ques in the end - diff b/w outliers and leverage points, how to detect leverage points, when is a point called an influential point, how do you determine p,d,q parameters for an ARIMA model (without autoarima).

Round 2 - 16 ppl + 4ppl waitlist were selected. Broad division with one panel focusing on technical and one on business aspects. My round 2 was with the technical panel. Rate your

coding skills (answered a high number).

1. Write code to find the sum of first n fibonacci numbers. How can you optimize this?
2. Write code to find the number of primes less than n .
3. Given a paper with 5 regression proofs and asked to choose. (Chosen ques - prove $y(\text{cap}) = p^*y$), p is symmetric, p is idempotent. p is the projection matrix. Answered every ques.
4. Ques on why I joined pgdba?

Round 3 - Almost all ppl who had round 2, also had round 3. This was supposed to be the business panel but it was late in the day and they were hurrying. Asked me 2 puzzles -

1. https://en.wikipedia.org/wiki/Two_envelopes_problem (could not answer)
2. A variant of find the 3 fastest horses puzzles from gfg

Deepak Ganta

1. Favorite project
2. Time series asked about Covid affect and outlier detections
3. Statistical tests that happen behind the scenes of ADF one structural test
4. How to model level shift time series
5. One puzzle related to some auction prices. What will be the approach if you have to bid for a picture of price unknown but u will know if u sell it you will get $1.5 \times \text{price}$ of picture. how to do
6. Why Regression can't be used for logistic regression beyond bound between 0 and 1
7. PORs related to me
8. General hobbies

Round 1:

1. Basic Introduction was asked
2. What were the significant variables i got in my regression project was asked and why we got those as significant
3. In-depth questions were asked on the Time series project. Various questions were asked about exogenous variables and dummy variables were asked
4. How would we deal with a time series during the holiday periods and Christmas seasons was asked
5. Basic aptitude puzzle regarding netflix was asked

Round 2

1. Asked about the introduction and home city
2. Time series questions were asked about outliers and the effect of the Russia Ukraine war. How would you forecast if a war situation again arises
3. Some RTSM question was given to solve
4. Basic probability question on Cricket was asked. In a 7 match series during which match it's most probable that a team will win the series
5. Asked to write prime number detection pseudo code in an efficient manner and asked about the time complexity

Suggestions:

1. SAAPM course during the IIM semester helps a lot to answer time series questions
2. Try to practice basic pseudocodes during the FADML course in IIT Semester
3. For interview purposes always pick 2-3 exogenous variables to answer during the discussion of the time series project. So accordingly choose your project now

Anirudh

R1

1. Questions on my aluminium time series project
2. Exogenous variables for aluminium price prediction
3. What is PCA? Can you apply PCA on time series?
4. Guesstimate

R2

1. Time series project grilling
2. More grilling on exogenous variables
3. Parameter in XGBoost that I had used to handle class imbalance
4. Some question regarding multicollinearity(basic one)
5. Questions on inference: Plot size and power on a graph
6. Puzzle(wasn't able to answer, very tough)

Yoogesh Kumar M

1. Tell me about yourself and what kind of projects you have done? In the projects that you have done so far what was the thing you found while analysing the data. Explain the timeseries project
2. Why did you stop with 2007 (I had forecasted till 2007)
3. What model did you used and what was the final sarima order
4. How will you explain this inference from the model to the business person
5. Sarima you have order of (0,1,2) what does it explain
6. What are the exogenous variables that can be used
7. oil price time series data you have how will use it as an exogenous variable
8. Does an exogenous variable require to have stationarity
9. Explain the ADF test cases

10. what is ljung box test
11. How will ACF and PACF plot be for AR 1 process
12. what does the beta value represent in logistic regression
13. how would the vary the threshold values in classification value
14. now if you need to model to predict how long he will take to default what would you do
15. what the defaulters balanced
16. many data we will not have number of days he took to default so how will you appoch
17. in which model you will have high error
18. what is AIC and BIC
19. Do you have any questions

Sachin Goyal

PPT Round (40 mins):

(Our time in the case was about the regression, where we had to predict Residential property value index (RPVI) for the US counties. The Challenge was about finding input features that we include in the regression model)

1. There were 2 panelists in this round, the questions were asked after we described our solution to them. they asked basic questions about the regression model and explain about each feature that we selected and why?
2. Each feature importance in the final model and why did we take nonlinear terms of some features?
3. how did you apply Lasso and Ridge regression etc?
4. Did you Compare with XGBOOST or any other non-linear models as well?
5. Business Use case of the problem?

Round 1 (40 mins):

1. I was asked to explain any one of the projects in detail. (I explained regression, so most of the questions were from this project only)
2. Top 5 imp variables in the regression model, then there were questions on each variable meaning?
3. How did you handle nonlinearity in data?
4. What is Lasso and ridge regression and the difference between them. What does R square mean?
5. What is stepwise selection and why did you use AIC?
6. What is multicollinearity and its impact on regression coefficients.
7. What is CLT assumptions and for which distribution we can't apply CLT
8. Type of population and what does unbiased, consistent mean?
9. Basic questions in classification project, different metric we use for evaluation in classification? What is the Rouge-L score?
10. Why did you choose regression project only, and not recommendation or NLP project?

Utkarsh Attela

Pre-interview Round

1. Walk me through your resume
2. Why PGDBA?
3. What are the different types of missing values? How do you treat them?
4. What is SMOTE? Why sometimes SMOTE should not be used for handling class imbalance? What are the other types of sampling techniques for minority class? Should SMOTE be applied before or after the train-test split?

5. What are LSTMs and Transformers? What are the differences between them and what was the reason that we needed a transformer over LSTM?
6. Examples when precision/recall should be chosen as a metric for evaluating the performance of a model, explain why?
7. What is Groq hosting LLM? (was mentioned in the resume)
8. What are the assumptions of the Central Limit Theorem?
9. Is accuracy a good metric for PPMI clinical data (Parkinson's disease dataset – I did a project on this)? What other metrics could have been used and why would it have been better?
10. If we have two models which predict if an individual is suffering from a disease (like model 1 for cancer, and model 2 for HIV), can we use the same metrics to evaluate the performance of the model? If yes/no, why?
11. If the model says someone has cancer when they actually don't, how will you handle the consequences of this kind of mistake in the model? How do you think about the cost (impact) of giving a false positive result? How will that cost be incorporated into the model's decision-making?
12. What is SVD? Explain it's working for a sparse matrix

Niranjan Herwadkar

DCL Presentation round:

- Questions on variable selection and feature engineering
- Questions on residual diagnostics in our regression model

Pre-interview round:

1. Long discussion on the following problem: How to construct an index with low variance? I was asked to state the optimization equation for this problem. What if short selling is not allowed? How will the equation change?
2. Discussion on Beta. How do we estimate beta? How does the estimate of beta from regression differ from the estimate that we get from direct formula?
3. Long discussion on my Time Series project on Volatility modelling. A follow up question which was difficult for me to answer: Which model can we use for a time series that shows high regime switching behaviour?
4. Asked me if I knew quantile regression?

Final day round1:

1. Discussion on a Business Analytics case study – How to reduce churn for Netflix?
2. I was asked to show Type 1 & Type 2 errors in hypothesis testing by drawing the distributions of H_0 and H_1 .
3. Discussion on my SARIMA time series project.

Final day round 2:

- I was asked to write codes for these simple problems on paper:
- Fib(n)
- Check if n is prime
- Some proofs in Regression:
- Orthogonality of $\hat{y}$ and e
- Tests to check distribution:
- Is there any parametric test?

- Name all tests that you know for this purpose.
- Case study on Finance which I don't remember clearly.

Note: I was asked many questions related to Finance, probably because my resume mentions it.

Anushka Nayak

Presentation Round

- Asked to explain the steps followed in the project, including:
 - What was done
 - Why it was done
 - What outcomes were obtained
- Focus on:
 - Output variables
 - Polynomial regression
 - Variability explained
 - Use of Lasso and Ridge regularization

Piyus Pramanik

1. Questions about case that was given, went deep into different possibilities and choices
2. Cross questioning about different projects mentioned in the cv
3. Mainly questioned were concerned about deep understanding of linear regression

SUN PHARMA

Vikash Kumar Gupta

Round 1

1. Introduction : About work experience and role etc.
2. Why data science?
3. Which is your favorite project and explain:
It was a regression project on uber car price prediction,he had used polynomial transformations and interaction terms
4. Difference between lasso and ridge regression, and how do they help in combating multi collinearity?

Yoogesh Kumar M

1. Tell me about your self
2. How did your automation pipeline reduce 80% of the deployment pipeline in your work
3. Why Sun Pharma
4. The Novartis Case competition is done by many people. How big was the team, what is each team member's contribution?
5. What are the interesting things you learnt doing this case competition
6. What do you do during your free time?
7. What position do you play in Hockey
8. You have mentioned you worked on UAV during your college days what is this Flight safety check system how does it work??
9. Do you still miss your Electrical core day??

Suggestions:

1. Resume building -no loopholes in the project

2. brand spectrum of projects-Clustering, Time Series, Recommendation system-script for each project explain - detailed and brief
3. Mock Interviews
4. UDL, ISLR & Youtube videos
5. 100 days ML
6. DSA
7. end-to-end-Project

Kshitish Raj

1. Why did you leave Deloitte?Why did you take up a sales role after Deloitte?

He framed and defended that the sales role involved technical and analytical challenges

2. How would you increase revenue for Sun Pharma?

He had briefly seen on the internet about sun pharma, he had seen that in some aspects they were in 10th position, so he framed his answer corresponding to that aspect because there is room for development. He also discussed customer and market segmentation that can help in generating more profit.

3. What do you dislike about the PGDBA program?

He said having to shift from hostel to hostel even within the same institute.

4. How would you use data science to solve this problem?

He said , this is not really a data science problem as the solution is likely to involve some solution which is not dependent on finding deep patterns in data. Interviewers still pushed for data science framing.

5. Do you have any questions for us?

He said yes, how is Sun pharma handling the impact of Trump's tariff on India? They had a discussion on this, and this seemed to have impressed interviewers.

Harsha pabbaraju

1. Many questions about roles and responsibilities in work experience
2. Picked a project he had done for a company he had worked before (analytics project) and went deep into that project details about his contribution
3. Why did you switch from chemical engineering

Anupam das

1. Introduce yourself.
2. My previous work experience was at ZS I explained what I did.
3. The interview went on a different tangent from there. They started questioning me about US speciality generics market
4. What is the business model of PBMs?
5. Differences between Indian and Us pharma market
6. This wasn't very technical; they didn't ask anything regarding my projects

Prashant bisht

1. They mostly asked about the workex. There was no difficulty in answering the technical part because they know nothing about ML. Project related they will ask, 1-2 questions. Easy level.
2. If you have mentioned anything like you did a savings of some lakhs, increased efficiency by some % in CV, they will ask how did you reach these numbers. They will discuss the numbers in detail.
3. Mostly it's about the fit and vibe, if they like you, they take you. They will keep the interview very interactive, you will not feel any pressure.

Case comps: Not that much. But good to have 2-3 case comps in your CV. Winning case comps gives you confidence and that helps. Sunpharma asks to give an excel sheet of students with casecomps they have won and POR for shortlist. How well you know them is for sure important, but topics should be like you have solved some business problem, not just build a model on kaggle dataset. And there should be variety in topics.

Sai kiran

Sun Pharma- one round experience based, one round resume based- Projects & tasks

JPMC

Vikash Kumar Gupta

Test round

1. Basic aptitude questions ,Quant questions like puzzles and reasoning questions and DSA questions
2. Basic questions related to work-time mainly to check time management and accuracy
3. Difference between XG boost and Random forest

First round

Mainly questions about projects

1. Clustering project : What is Davies Bouldin score? Silhouette score?
2. Explain and differentiate lasso and ridge regression
3. Explain chi square tests, sharpe ratio
4. Rag project : Bert score, rouge 1 and 2 scores
5. Different evaluation metrics like F1 score, Ljung box, Durbin Watson test etc.
6. Case based question : What factors do you take into accounting to launch a product? How would you decide on pricing?

Second round

Questions about work experience, motivation for choosing this course and career path, where do you see yourself in 5 years

Coding question : Given an integer n, find the highest power of 10 that divides it, they insisted on coding explicitly not just idea

Anirudh

R1

1. Asked about my work experience(time series)
2. Asked my favorite project, again time series
3. ACF, PACF, AIC, BIC, stationarity, ARIMA components
4. My project on time series
5. Project on DL. What is transformer, attention, history of NLP.

R2:

1. Coding question 1 2 3 5 6 7 find missing number
2. Time series imputation
3. Questions on my work ex

Harsha pabbaraju

Test round: DSA there will be two questions and only if you do both , you will be selected for next round

First round: A rapid fire round of questions mainly centered around the projects and resume 15-20 questions in 15 mins. Knowing the project well is important

Second round

1. Basic coding questions in python and sql
2. Write a program to find the number of os in n!
3. SQL join related questions
4. Window function related questions

MASTER CARD

Utkarsh Attela

Round 1

1. Introduce yourself.
2. Which type of music do you like to listen to?
3. What is SVD? Explain how it works and how it is different from PCA.
4. What are eigenvalues and eigenvectors?
5. Why is standardization required before applying PCA?
6. What is the rank of a matrix? Explain its importance and how it is used in SVD.
7. What is a convex function? How is it represented as $f(x)$?
8. What is gradient descent?
9. If a new feature is added to Netflix and deployed, how would you determine whether it is performing better or worse for the company?
10. When deploying the new feature, should it be released to all users at once? Why or why not?
11. What is a confidence interval, and how is it different from a prediction interval?

Round 2

1. Introduce yourself.
2. Explain one of your projects in detail.
3. (Follow-up questions may be asked based on the project.)
4. What is cointegration?
5. What is R^2 , and what does a high value mean?
6. What is multicollinearity, and how can it be handled?
7. What is the difference between CatBoost and XGBoost?
8. What are the hyperparameters in XGBoost?
9. How are missing values treated in an XGBoost model?

10. What is dropout? What problems can arise because of dropout, and how can they be solved?
11. Do the dropped nodes remain inactive during testing? How do they contribute when making predictions for a new observation?
12. What is the vanishing gradient problem, and why does it occur? While training a model, how can we identify that it is suffering from this issue?
13. What is the difference between batch normalization and layer normalization? Where is each used, and why?
14. Explain the encoder–decoder architecture of a Transformer.

Round 3

1. Introduce yourself briefly
2. Why PGDBA after 3 years of work experience in your previous company?
3. What do you think have improved before and after joining PGDBA?
4. Why Mastercard?

MASTERCARD AIG

ROUND 1 (eliminated)

- Equations of Lasso and ridge regularizations, more about regularizations
- XGBoost
- Random Forest
- Explanation of my favourite project (recommender system). Asked me to design a recommender system which has Logistic Regression as its final model.
- Why low CGPA in B.Tech?

Anushka Nayak

Round 1

- Focused heavily on PCA and SVD, asking questions in significant depth:
 - Mathematical and conceptual fundamentals of PCA
- Questions on:
 - Handling dropout values
 - How change testing was done and how values were dropped
- ANN-related questions:
 - Libraries used for training (PyTorch, TensorFlow)
 - This was unexpected given the lack of coding focus
- Some project-specific questions were also asked

Vasu mavani

1. Explain your favourite project
2. Related to project - what is limitations of knn imputation?
3. How knn works in general?
4. Assumptions of linear regression
5. Cauchy schwartz inequality
6. Kolmogorov Smirnov test
7. How to check correlation between categorical variables?
8. Coding problem: Find the kth largest number from a list

SWIGGY

Vasu Mavani

Round 1: (Coding round)

1. Given 2 sorted lists, how to merge it into a single sorted list.
2. SQL - Department wise top 2 employees based on salary
3. Implementation of simple linear regression with numpy

Round 2: (ML Round)

1. Explain one of the project
2. What are different classification measures apart from precision, recall, accuracy and f1 score? Questions from logistic regression Why could linear regression not be applied for classification problems?
3. Explain decision tree
4. Explain MLP (Multi layer perception) classifier
5. 6 chances, probability of hitting target 0.25, what will be the probability of hitting the target more than at least once?

Round 3: HM round

1. Search in the Swiggy, like piz then it shows suggestions, how it works?
2. Regular expression I have told with some patterns
3. Then she mentioned what about the scale
4. Then, Swiggy delivery boy wears a swiggy t shirt and uses Swiggy box only to deliver, what impact does it make overall and impact the customer's mind?
5. What kind of problem is it?
6. Which ML/DL technique can be used to solve this?
7. What will be the different set of output it will have?
8. What will be the loss function?

9. What other features you will use for identify this apart from image? Like delivery boy behaviour (Not from the image)
10. How to make it mandatory for delivery boys to wear the t-shirt?
11. How could a penalty like 10 rs of not wearing a tshirt can be included in the loss function?
12. How based on the model only we can penalize the delivery boy like low threshold (when model says not wearing, high chance of not actually wearing it - false positives and false negatives) or frequently flagged delivery boys (repeatedly not wearing t-shirts according to model)

Prabhanjan

Round 1: Coding three questions, First on python basics , Second on linear regression from scratch using numpy, Third DSA

PNG

Piyus Pramanik

Online assessment test

Behavioral questions, quick calculations and connecting dots like puzzles which test short term memory to shortlist candidates.

First round

1. Introduce yourself
2. Which is your favorite project? Describe it : For every part of the project description they cross question so understanding the project is of immense importance, He chose market basket analysis about clustering.
3. How do you evaluate the performance and correctness of your work.

Second round

1. Situation based questions
2. Give an instance of a situation you faced and how did you handle it?
3. They ask about core values of P&G and the candidate is expected to answer questions and situations centered around it.
4. Give an instance from previous work experience where you took responsibility not allocated to you in which you gave good output.
5. Give an instance of the situation where you were not ready to handle, but you still managed to give a good output
6. You are a senior data scientist at P&G working on a project with 15 stakeholders. Fourteen support the project, but one key stakeholder—the CIO—is not convinced. Even if your own team is also hesitant, how would you proceed? Explain how you would handle this situation, focusing on values and integrity rather than authority or consensus.

Third round

1. Questions to judge history and behavior and value of the candidate broadly.
2. Tell me something about yourself not in the CV

Purpose of this round was to test organisational fitness of the candidate

Vikash Kumar Gupta

First round

1. How to normalise data, different cleaning methods and imputation methods and basic questions from different models

Second round

Case based question after seeing cricket as hobby in the CV:

It was a CAR type question (context, action and results)

Case : BCCI hired you as their first ever data scientist, what exactly are you going to do to increase winning percentage most effectively? What data you need, how to collect those and how would you use them?

Prashant Bisht

Round 1:

Someone from singapore office, her accent was not very clear, and she said to only answer if you know and don't fool around (something like that)

1. How will you explain bias variance tradeoff to a layman?
2. What is CI/CD pipeline? Only told fullform and she was ok.
3. why some models fail when deployed in production?
4. ML questions but basic level.
5. prepare some general model deployment questions.

Round 2:

Same questions as mentioned in batch 9 png interview.

How will you handle a project associated with multiple departments? Only management related questions. o technical. Prepare a framework to answer them. You will have a better chance.

Anupam das

Round 1

Very technical. If you have a good grasp of technical concepts mostly ML and basic DL is required. The interviewer asked to refrain using Gen ai or llms for any of the answers.

1. Started with linear regression in depth questions about assumptions what happens when they are violated how to test them
2. What is ridge, lasso
3. Bias variance tradeoff

4. How is decision tree and rf different.
5. How will you check model performance whether it's working well or not after deployment
6. What do you use for deployment notebook file or .py files
7. How does XGBOOST work.
8. What are the few hyperparameters for random forest.
9. What is pruning?

Round 2

Situation based round

Question 1

Say you are working on 4 projects one takes 2 weeks another 3 weeks another 1 month and another 2 months

CFO now tasks you suddenly with a very critical task how will you proceed.

Question 2

Say you are presented with a new opportunity but learning the required skills for it will take 1 year or more to fully grasp it. The opportunity is going to commence in 5-6 months how will you proceed.

Round 3

This was again a situation round mostly to test leadership skills

Tell me a situation where you lead. I answered but it was a situation of solo leading.

The interviewer wasn't satisfied with my answer and asked me to tell a situation where I lead with others.

Be strong in technical and go through past interview experiences for situations and leadership round.

Kenvue

Vikash Kumar Gupta

Group discussion round:

1. Can AI replace data scientists?
2. What is the most important role of data science in supply chains? If it is demand forecasting, why can't we automate that?

First round

1. RAG project based questions: How did you evaluate performance?
2. What are lexical similarity and semantic similarity?
3. Clear in detail mathematical explanation of PCA and eigen values and eigen vectors

Second round

1. Work ex related questions, was you previous work similar to cricket ground or football ground and why?

Anupam Das

1. Introduce yourself
2. Questions on linear regression
3. Ridge lasso, multicollinearity.
4. What is VIF?
5. General questions on multiclass classification
6. How will you do multiclass classification using only a binary classifier.
7. I had applied a lot of different methods in my regression project. The interviewer asked- if someone doesn't have that much time to apply everything how will you check.

8. Then questions on my extracurricular work in trilytics
9. Then asked about my hobbies
10. Tell me about one difficult decision you had to take?
11. What kind of work do you like

Sai kiran

GD 6 members 4 selected(Can AI replace Data Scientist!)

8 members shortlisted: Case study study- Recommendation System, Regression
Project-use case

BCG

Sai Kiran

Test: You were allowed to use documentations of pandas, scikit learn etc. No AI apps were allowed

2 probability questions based on conditional probability (Medium to Hard level)

7 ML based multi choice questions (multiple answers were allowed and 5-6 options were there)-Most of them are based on hypothesis testing, linear regression and logistic regression (Easy to Medium level)

3 questions each for 160 marks- Goal is to check your pandas (data manipulations and cleaning) and modelling skills

In 1st question one table was divided into 4 subparts and there are 3 other different tables. For this question you need to be aware with pandas functions. In 2nd question again you will be given one dataset on that you need to do further cleaning. In 3rd question you will be asked to apply model on cleaned data where train, val and test datasets were given and kept some minimum rmse to pass the test cases. I was given regression question

I have applied for Risk profile

Round1:

Started with my regression project, asked about

- 1) What is the project about?
- 2) What are the features you have considered?
- 3) What is meant by lasso and ridge?
- 4) What is the difference between lasso and ridge?
- 5) How lasso and ridge work?
- 6) What is meant by regularization?
- 7) Why do we add penalty to loss function for regularization?
- 8) What are the assumptions of linear regression?

I have credit risk modelling as one of my projects (I would highly recommend you do this if you are interested in risk profile)

- Asked me to explain what all I did in credit risk modelling? (about dataset and variables considered)
- How did you do the feature selection?
- What is Weight of evidence and Information Value?
- What is LGD?
- For a given variable asked WOE will be higher or lower?

I have Fraud detection project as one of my projects (I would recommend you do any fraud related project if you are interested in risk profile)- I have used LSTM, Random Forests and XGBoost

- Asked me to explain about project and variables considered
- What is XGBoost and how it works
- What's the difference between XGBoost and Random Forests other than bias variance aspect

Round 2:

- Why risk?
- Why PGDBA?

From Fraud detection project

- Asked me to explain about the dataset and what all features were considered?
- Actually, the data is sequential, so he moved to time series asked me to explain the steps involved in timeseries project like from start to end and what all tests you used?
- What is stationarity?

- What will happen to predictions if we use non stationary data without making it stationary?
- What is heteroskedasticity and what are the issues?
- You were given 2 minutes time and you need to explain about your project to CEO of the company. How will you explain?
- What is ROC and how will you explain this to CEO

Round 3:

- Asked about my family and schooling
- Why risk?
- Why PGDBA?
- What all types of regression you know?
- How many loss functions you know and list them out?
- Explain AIC to 8th class student
- Explain P-value to 8th class student
- What's the one thing that you want to change in you?
- How you want to communicate with client apart from written and

Yoogesh Kumar M

Round1:

1. Tell me about your self
2. Which is the ML Model you know well about. (Ans:XGBoost)
3. What are the hyperparameter used to prune the XGBoost Tree
4. Can you use XGBoost for forecasting in time series data?
5. How well you know Statistics rate your self in the scale of 10.
6. You are given two real number series what kind of statistical test be performed
(Mean, Variance, Distribution)

7. What is the test used for Distribution?
8. Where else we use KL Divergence.

Round 2:

1. Explain Your Anime Recommender Project
2. Why did you use Two Tower model
3. How did you use Graph Neural Network in recommender
4. Case:
Use client is a Bank and they are asking to build an interactive AI voice chat bot to attend the customers grievances.
5. 3 Strengths and 3 Weakness

Round 3:

Case: Build a model to predict loan defaulters.

Prabhanjan

Round 1

1. Intro
2. Workex questions, basically what I did there
3. What new things you would have suggested in your previous company based on what you have learnt in pgdba
4. End to end process for sentiment analysis in detail.
5. Linear Regression Assumptions
6. Questions on regularisation.

More of a stress interview

Harsha pabaraju

Test round

Proper hands on coding questions . Like pandas, combine different files into one dataframe and manipulate values.

First round

Cross questions about projects about all sorts of decisions made, why this why not this kind of questions, so having clear grasp of every project is very important

Second round

It was an elimination round, a lot of cross questions again about projects and some technical questions, nothing special.

POLYCAB

Vasu Mavani

- Health related concerns?
- Temperature and humidity?
- If new projects after 3 months new requirements what will do?
- Planning to do business after 3-4 years?
- Everything related to work experience in GenAI?
- Hallucination and how to control it?
- Chainlit and react have you built it?
- Adaptive to learning new technology?
- How to address hallucinations?
- What is embedding?
- How would you explain it to 10 year old child.
- Interested more in the UI part How to check the ROI of the project
- What do you know about Polycab?

Adwaith Aravind A: Totally related to work experience

Prabhanjan: Full discussion on Workex. What I did, challenges, tech stack, end to end process details etc.

Pirama Finance

Prashant Bisht

PF selected us based on our CV. What I felt based on the people selected for the interview, they have 2 projects in common. 1 classification(Finance fraud/loan default) and other RAG projects. I had both of them in my CV. There were 2 panels. Each panel had 2 persons. One asked related to LLM's and RAG. And others asked about traditional ML related to classification.

Since there are 2 teams in PF one is the credit risk modelling team, so classification. And others are the GenAI team, LLM and RAG.

1st interviewer:

1. Explain how the pre-training is done in BERT? He was expecting “masked language modelling” and “next sentence prediction”. Then counter question:
2. How BERT does classification? He was expecting BERT architecture. CLS token, probability distribution over entire vocabulary using softmax. He asked BERT since I had mentioned BERT multiple times in my CV.
3. What is SVD? How is it used in the recommendation model? Since I had a project on movie recommendation using SVD. Prepare the meaning of each matrix also. $A = U \sigma V^T$. What is U, V and sigma?
4. Explain your RAG project.
5. Asked about the RAG architecture and the highlighted terms of CV in RAG project, like type of retriever I used like MMR. Why this retriever you used. What if 5. The query is ambiguous then which retriever to use.

This was a complete technical interview, all questions asked were CV based.

The 2nd interviewer asked about the ML questions.

1. The model is performing well on the training set but not on the test set. Why?

2. Suppose a model is trained on 3 years data, let's say 2017-2020. It performed well on the 2021 data. But fails to perform well on the 2022 data. Why? (Prepare this well with an example)
3. How will you explain this model not performing well on 2022 data to a business person?
4. There is a classification model that is giving good ROC-AUC value. But the model performance is not good. Why? ROC-AUC might not be the best metric for the model. if there is class imbalance then ROC-AUC is unreliable, PR-AUC should be checked.
5. He countered that, if roc-auc is good, shouldn't pr-auc also be good?

I didn't feel at any point the interviews went wrong technically.

Sarthak Sablania

Mostly Technical questions about general concepts of ML- Classification Methods, focussed on projects, recommendation systems, Regressions and SVD, few scenario questions on ML- such as training testing, model works on certain type of data why not on this data type, workex and cv related questions- so be thorough with whatever is being mentioned

Ishu Patel

- Introduce yourself.
- Discuss your regression project.
- How many variables were there?
- Can you name some of the variables?
- Why did you choose certain variables and not others?
- Why didn't you use XGBoost Regressor in your project?
- How would you convince company management that your model is reliable, assuming they don't understand any technical jargon?
- What is the Jackknife method?
- How would you verify that a suspicious value in your dataset is not just a wrong entry?
- Since we provide loans for cars and your regression project is on car price prediction, how can we use your model in our business context?
- Discuss your RAG (Retrieval-Augmented Generation) project.
- What were the components of your model?
- Suppose we want to use this model in a specific field (say, finance). Which part of the model would you need to update or change?
- Suppose we have Aadhaar data (both PDFs and images): some are full-size forms, and some only contain name and address information. Using vector embeddings, how would you separate the full-length Aadhaar PDFs from the smaller Aadhaar card files?
- Do you have any questions for us?

Anupam Das

There was only 1 round. They were very interested in LLM, Rag projects and DL

- They started off with my agentic ai project
- Explain the pipeline. What is docker streamlit?
- Asked about the limitations of my project can a 1 hr video be summarised if your model has only a 15 min context window.
- Then the key terms what is bert t5 differences.
- What will you use to deploy a mail classification task
- Ask about llm fine tuning.
- What are optimizers in DL?
- What are skip connections
- What is vanishing gradient problem how to handle it.
- What is TF-IDF?

Adwaith Aravind A

1. Discussions about projects and basic questions about regression
2. What are transformers and what is the principle behind it?
3. Explain the RAG project you have done : How did you build it and evaluated it?
4. Discussion about a credit risk project he had done, how did you remove some features? What are information value and risk based scores?

Utkarsh Attela

Round 1

1. Explain the Legal Document Assistant – RAG project in detail. (Follow-up questions related to embeddings, text chunking, and FAISS vector database.)
2. Which API did you use?
3. Explain the encoder–decoder architecture.
4. BERT and Flan-T5 Small (what type of models are they?)
5. How are embeddings created in BERT? What is the difference between BERT-generated embeddings and GloVe embeddings?
6. What is FAISS?
7. Suppose you have 500 features, and you need to perform a binary classification task. How would you proceed?

Anushka Nayak

Round 1

- Introduction
- Discussion of two projects:
 1. Agentic AI project
 2. Credit risk project

OLIVER WYMAN

Ishu Patel

Round1

- Introduce yourself.
- Explain your regression project.
- What are the variables used in your regression project?
- What are the assumptions of linear regression?
- What is multicollinearity?
- What will happen if the normality condition is not met (in terms of biasedness and consistency of estimators)?
- Explain Ridge and Lasso regression their loss functions as well.
- What are R^2 and Adjusted R^2 ? Why do we use them, and when do we use them?
- Case: How would you estimate the sales of Apple products in India? What variables would you choose and why? What feature engineering would you perform, and which model would you use?

Puzzles:

- Three ants are sitting at the three corners of an equilateral triangle. At the same time, each ant randomly chooses a direction and starts walking along the edge of the triangle, either clockwise or counterclockwise. What is the probability that at least two ants collide?
- A drawer contains 10 white socks and 10 black socks. If you pick socks at random without looking, what is the minimum number of socks you must draw to be certain of getting a matching pair of the same colour?

Round2

- Introduce yourself.

- Are you comfortable with Python?
- Is a dictionary ordered or unordered in Python?
- Write code to convert an unordered dictionary into an ordered one.
- Case Study: An office supplies company is losing clients. What variables would you consider to identify possible reasons?
- Guesstimate: Estimate the number of tennis balls that can fit inside a passenger plane.
- HR Questions:
- What motivated you to switch your career from your core background to analytics?
- Suppose in your work you made a mistake, and you know that if you don't tell anyone, nobody will ever find out. What would you do in this situation?

Round3

- Introduce yourself.
- What is interesting about Rewa (My hometown)? What is the economic model of Rewa?
- Pick the most interesting project from your CV and explain it in detail. (Some cross-questions around the project.)
- Explain your Generative AI project in detail. (Expect a few cross-questions around it.)
- What is Agentic AI? Tell me something about it.
- Case Study: Suppose you are given a business problem.
- What variables would you choose and why?
- What feature engineering steps would you perform?
- What model would you finally choose and why?

- Case: (The exact case I forgot, but it was mainly about which variables you would choose and why, what feature engineering you would do, and finally which model you would choose.)
- Puzzle: A simple puzzle to calculate the area between two geometries.
- Do you have any questions for me?

Round4 (Partner round)

- Introduce yourself.
- Calculate the total number of trailing zeros in 1000!
- Case: Suppose you are the owner of the Delhi Daredevils team and want to purchase Virat Kohli in an auction. What valuation would you decide for him?
- Do you have any questions for me?

VASU Mavani

Round-1

- Explain one of the project
- How to decide which evaluation measure is better?
- What is stationary?
- Difference between strong and weak stationary?
- Difference between df and adf test
- Assumptions of linear regression
- Why is stationary required? What problems will you face if not stationary?
- Why should residuals be normally distributed? If not, what problems will you face?
- What is multicollinearity? What problems will you face if multicollinearity is not removed? Isn't multicollinearity and heteroscedasticity the same?

Utkarsh Attela

Round 1

1. Walk me through your resume
2. Explain one of the machine learning projects you worked on in your previous company.
 - I. What features did you use, and why?
 - II. How did you determine which features were important?
 - III. Were all features selected based only on statistical criteria, or did you consider domain knowledge as well?
 - IV. Why did you choose Random Forest for modelling?
 - V. What challenges did you face during deployment, and how did you handle them?
 - VI. Were there instances where a feature was initially relevant but became irrelevant after deployment (or vice-versa)? How did you monitor and detect that?
1. If you are using a dataset with missing values and building two models, one is Linear Regression and other is Logistic Regression, how are missing values handled in each? Do we need to impute missing values for both? Explain how missing values are treated in XGBoost.

Round 2

1. Introduce yourself.
2. Questions related to work experience.
3. What projects did you work on in the previous year? What challenges did you face, and how did you tackle them?
4. Conflict-based questions:

5. If there are only two people on a project (you and one more team member) and there is a disagreement while building a model, how would you handle it?
6. If one of your team members makes a mistake in the project, how will you handle it? Would you inform your manager or sort it on your own without informing your manager?
7. If you make a mistake in a deployed project, and no one except you is aware of it, what would you do? Would you admit your mistake?

Prabhanjan

Round 1

1. Intro
2. Details of linear Regression project, Assumptions etc
3. Simple puzzle - check geeks for geeks
4. Question on probability - x red socks y black socks etc.
5. Case study- Xiaomi wants to find demand for a mid segment phone in China. Whats your approach? More of an analytical case study

Adwaith Aravind A

First round

1. Linear regression related questions: what is VIF, how is it related to R^2 , how to use these in feature selections, what are the regularisation methods you know? What is multicollinearity and how lasso and ridge regression methods help in that case?,
2. What is R^2
3. If you have 200 features, how do you select the best features and models ?

4. About multicollinearity, what are the common thresholds for vif value and how to address it?
5. Explain XG boost, cat boost and light bgm, which of these require one hot encoding? How do these models handle categorical features
6. Bagging and boosting (difference between them)

Second round

1. Questions about extra curricular interests, why are you interested in consulting?
2. Pick a project you have done and explain its business impact : He took the credit related project about insurance dataset , and they discussed about segmenting customers and optimisation

Anushka Nayak

Introduction, Core Data Science Questions

- Fraud detection
- NLP and classification
- Class imbalance:
 - What it is
 - Undersampling
 - Other handling methods: Oversampling, SMOTE
 - How SMOTE works
- Metrics and models:
 - F1 score
 - TF-IDF
 - Logistic regression:
 - How it works

- Use cases

Case Study

- Build a movie recommendation system
- Given features, asked how to design the system
- Detailed discussion due to prior experience in recommendation systems

Puzzle: Classic 100 switches problem

Credit Risk Project

- Full workflow explanation
- Emphasis on keeping projects business-oriented
- Feature discussion
- Statistical tests:
 - Chi-square
 - ANOVA
 - VIF (conceptual explanations)

Coding

- 2D array problem
- Counting capital letters in arrays
- String array manipulation
- Python basics

Round 2

- Introduction
- Coding:
 - Nested loops

- Time complexity
- Dictionary-based questions
- Credit risk project explained again
- HR questions:
 - Why Oliver Wyman
 - Why consulting

Round 3

- Presentation-based round
- Evaluated on four core principles (logos-based)
- Interviewer: India Analytics Head

Discussion Points

- Introduction
- Reason for leaving startup
- Work experience
- Credit risk project (again, repeated questions)
- Motivation for joining Oliver Wyman

RENew

Utkarsh Attela

Round 1

1. Explain the Legal Document Assistant (RAG project) in detail.(Follow-up questions related to embedding methods, text chunking, and the FAISS vector database.)
2. What is R_Square ? What is multicollinearity? Explain the need for Lasso and Ridge.

Round 2

1. Discussion on the same RAG Project.
2. Explain encoder and decoder.
3. Do you know what work Renew does?
4. How are these embeddings created? and how to improve if the model is not performing well, what are the likely problems that could occur.
5. What are word2vec embeddings and how is it different from Glove embeddings?

Round 3

1. What were the learnings of your previous company that you could apply in PGDBA
2. What were your key learnings from this program

Anupam Das

Started off my regression project. Mostly asking about the business aspects of it.

My project was on insurance charges prediction

If you have a dataset on health and you want to know the smoker status of people what will you do?

What is bert and t5. Can you use the encoder from bert and connect it with just the decoder from t5?

What will you do first to remove outliers or impute? Think in terms of train and test data

Overall if you're presenting a solution think about the deployment and large scale aspects of it as well.

Prabhanjan

Round 1

1. Intro
2. Detailed discussion on workex
3. I had two projects one on CNN and other on Predictive maintenance both relevant to their company. So detailed discussion on those

AUXO AI

Sushrut Joshi

Interview one: (2 guys from data science team)

Q: How many years of work experience do you have? What did you do at your job? And some other questions related to workex.

A: workex 3 yr, analytics background etc etc...

Q: Analytics case study:- given many factors around farming of berries, also gave multiple hierarchies for time series forecasting (example: region, farmer, time etc). decide 1) at what level to forecast, 2) how to forecast. Then got in depth of this as why you chose this approach, what would happen if you change some things etc.

A: I fumbled a bit, but they will help if you try telling your approach. They just checked whether we could approach the problem in a structured manner or not (which I think I was able to :))

Q: basic DL/ NLP questions in a rapid fire manner: transformers, finetuning, attention-self attention models, etc.

A: One should be thorough with all these concepts.

Q: Do you use LLMs in your projects? Which models did you use?

A: I thought this was a trick question, but I answered it honestly. Basically at work they call all your LLMs to do majority work and what I sensed was he wanted to know which models I use and for which tasks. With that I mentioned Agentic mode of chat GPT (just to give him a hook).

Q: Do you know about agentic AI?

A: Yes, I told what I read the night before...

Q: Do you know about RAG?

A: told whatever I read the night before...

Q: follow up questions on RAG: why do we use it? Why can't we feed everything to LLM and get answers? How do we retrieve the data?

A: One can answer these questions by giving a 3-4 hr read about RAG.
(I didn't have any project on RAG, but as it was mentioned in JDF, I read about the stuff).

Q: coding question: simple data manipulation, they gave me a small dataset in json, asked me to load it, filter it, merge, groupby etc.

A: done

Second interview: (2 interviewers- ML part was kind of rapid fire for around 40 min)

Q: basic introduction.

A: normal prepared answer.

Q: lets test your traditional ML knowledge, what is regression? When not to use this? Difference between SLR and polynomial regression? When do we use polynomial regression? What is logistic regression? How can we do the multi class classification using logistic regression?

A: answered all the questions and the interviewer seemed satisfied.

Q: Do you know about SVM? What are support vectors? What are different margins? How does changing the hyper parameter in the SVM equation change the classification?

A: answered all the questions and the interviewer seemed satisfied.

Q: Questions with respect to classification, how do we choose the number of clusters in k means? Explain elbow method in short? What is there on y- axis during the elbow method? Which clustering do we use in which case? (K-means, hierarchical, DBSCAN). Limitations of each methods?

A: answered all the questions and the interviewer seemed satisfied.

Q: Are you comfortable with Bayesian ML techniques?

A: I know where to use it and how to use it, but I'm not comfortable with the equation and stuff. (He then didn't ask on this topic)

Q: Are you comfortable with boosting algorithms? (yes...). How does the XGBoost algorithm work? Difference between XGBoost and CatBoost?

A: answered all the questions and the interviewer seemed satisfied.

Q: Another guy asked a coding question (based on hashing):- given a string, you have to give output of the letter which gets repeated first. Example: "swish" → s is repeated first, hence output is "s". "saasw" → here a is repeated first hence output is "a". "halo" → none is repeated, hence output is "None".

A: coded on jupyter notebook (screen shared). Used hashing for this.
follow up questions on code, is it optimised? What is the time complexity of this code?

(they try to look at your approach, if you are not confident with syntax, you can ask them if you can use the internet for syntax, they may allow, discuss the approach first).

Third Interview: (India head- cultural fit)

Normal interaction with India head, discussed things about location, company objectives, policy, joining date etc.

He indirectly mentioned that they need people for full time and are hiring for full time and not intern(basically giving PPO hints on the spot)

Note: I had given nearly every company's interview (almost all rounds), Auxo's interview was one of the most comprehensive interviews covering almost all the topics. This might be due to the fact that they mostly give ppo on the spot.

Decision Point

Prabhanjan

Round 1

1. Questions on each and every highlighted word in CV. What it is, mathematical equation, how you will explain to client/business person etc

Round 2

1. Detailed discussion on a project in CV
2. Workex discussion
3. Guestimate - Total runs scored in a year in cricket

Adwaith Aravind A

Very basic questions and a guestimate question

L & T finance

Prabhanjan

Round 1

1. Intro
2. Workex
3. Linear Regression Assumptions from project
4. Detailed discussion on A/B testing.

Round 2

1. What is skewness
2. What will it be for exponential distribution?
3. There is a normal distribution with its mean following exponential distribution. What will be the distribution of the the posterior
4. Asked to share screen and code logistic regression, gpt was allowed. Finally asked to interpret the results

Adwaith Aravind A

Round 1

1. Discussion about every project and cross questions about concepts involved
2. Detailed explanation of K-means clustering algorithm: Why to standardise features? How do you choose k?
3. Credit risk related terms explanation , what is “days past due”? What are the criteria for defaulting?
4. Questions about how would you personally choose a credit card? When are you willing to pay a premium?

Round 2

1. Asked about proficiency in python and asked to generate a vector and some basic plots
2. Asked to choose a concept (chose linear regression) : asked assumption, how to check linearity, why is it called linear regression.
3. Asked about logistic regression, log odds, sigmoid function etc.
4. Properties of linear functions, and asked to verify linearity of different functions given
5. HR questions about previous work experience.

Decimal Point

Prabhanjan: Completely HR type. Workex, leadership qualities, failure scenarios, lead team, etc only one round

Adwaith Aravind A: Totally related to work experience

JSW

Prabhanjan

1. Workex discussion.
2. I have written Business Intelligence in my CV. Questions on how a CXO dashboard looks. How is it different from any other dashboard? What metrics are important etc